TINITIATE
TECHNOLOGIES

TINITIATE · AI DEVELOPER MASTERCLASS

THE EVOLUTION OF AI

From deterministic rules to autonomous, multi-agent reasoning

01 AUTOMATION
02 TRAD. ML
03 GEN AI
04 AGENTS
05 LOOPS
06 FUTURE

- **Rules → Learning**
  automation and traditional ML
- **Learning → Creating**
  generative AI and LLMs
- **Creating → Acting**
  agents, loops and what comes next

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06 ERAS | 60+ YEARS | 01 JOURNEY

01 AUTOMATION | 02 TRAD. ML | 03 GEN AI | 04 AGENTS | 05 LOOPS | 06 FUTURE

ERA 01 / 06

Automation

1960s → 2010s

Automation: “If this, then that.”

The first step toward intelligent systems. Humans write every rule; the machine follows them perfectly, at speed, every time. Same input → same output.

HOW IT WORKS

Trigger	›	Rules	›	Action	›	Log
file, schedule, event		if / else, lookups		update, send, post		audit trail

FROM SCRIPTS TO HYPERAUTOMATION

- **1960s–80s** batch jobs, JCL, cron
- **1990s** shell & Perl scripts, macros
- **2000s** workflow & BPM engines
- **2010s** RPA bots that click like humans
- **2020s** hyperautomation: RPA + APIs + AI

EXAMPLE · ACCOUNTS-PAYABLE BOT

- Watches the invoices mailbox
- Extracts vendor, amount, PO number
- Matches against the purchase order
- Posts to ERP or routes exceptions

THE AUTOMATION TOOLBOX

Python | PowerShell | cron | Apache Airflow | UiPath | Power Automate | Zapier | n8n | Ansible

IS IT WORTH AUTOMATING?

- **Yes** if high volume, repetitive and rule-based
- **Yes** if the inputs are structured and stable
- **No** if every case needs judgment

LIMIT → NEXT

Rules can’t handle what they’ve never seen. A new invoice layout breaks the bot, so we need systems that **learn** the rules from data.

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01 AUTOMATION · 02 TRAD. ML · 03 GEN AI · 04 AGENTS · 05 LOOPS · 06 FUTURE

ERA 02 / 06

Traditional ML

1990s → 2010s

Traditional ML: learn the rules from data

Instead of writing rules, we show the algorithm thousands of labelled examples and it finds the pattern. The output is a **prediction** with a confidence score, not a fixed answer.

THE ML LIFECYCLE

Data	Features	Train	Evaluate	Deploy
collect & clean	engineer signals	fit the model	accuracy, F1, RMSE	serve & monitor

THREE WAYS TO LEARN

- **Supervised**: labelled data (regression, classification)
- **Unsupervised**: find structure (clustering, anomalies)
- **Reinforcement**: learn by reward (games, robotics)

CLASSIC ALGORITHMS

- Linear & logistic regression
- Decision trees, Random Forest, XGBoost
- k-Means, SVM, Naive Bayes
- Early neural networks

ACROSS DOMAINS

- **Finance**: card-fraud scoring
- **Retail**: demand forecasting
- **Real estate**: price estimation
- **Social media**: feed ranking
- **Logistics**: delivery ETA

THE ML STACK

Python · pandas · Polars · NumPy · scikit-learn · XGBoost · Jupyter · MLflow · SageMaker

LIMIT → NEXT

ML predicts but cannot **create**, and every task needs its own model and features. Deep learning on internet-scale data opens the door to generation.

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ERA 03 / 06

Generative AI

2017 → today

Generative AI: from inspector to creator

Traditional ML inspects data and labels it. GenAI creates something new: text, code, images, audio. Foundation models learn from huge amounts of data, then follow plain-language prompts.

FOUR ERAS OF LLM HISTORY

2017	2018–21	2022–23	2024+
Transformer: "Attention Is All You Need"	BERT, GPT-2/3: scale wins	ChatGPT, Claude: chat goes mainstream	multimodal, reasoning, open weights

CORE CONCEPTS

- **Tokens & context window**: how much the model reads
- **Temperature**: creativity vs. determinism
- **RAG**: open-book answers from your own data
- **MCP**: one standard port to plug in tools
- **MoE**: sparse experts, cheaper inference

THE 4 LAYERS OF THE GENAI STACK

- **Models**: Claude, GPT, Gemini, Llama, Mistral
- **Cloud**: AWS Bedrock, Azure AI Foundry, Vertex AI
- **Hubs**: Hugging Face, Ollama
- **SDKs**: Anthropic SDK, Google GenAI SDK, LangChain

ACROSS DOMAINS

Social: post & caption writer | Finance: earnings-call summaries | Real estate: listing descriptions | Retail: product Q&A bot | Logistics: shipping-document extraction | Dev: code assistant

LIMIT → NEXT

A chatbot answers once and waits. It cannot book, query, or fix anything itself. Giving the model **tools and a goal** turns it into an agent.

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01 AUTOMATION · 02 TRAD. ML · 03 GEN AI · 04 AGENTS · 05 LOOPS · 06 FUTURE

ERA 04 / 06

AI Agents

2023 → today

AI Agents: LLM + tools + memory + goal

An agent doesn't just reply. It decides what to do next, calls real tools (APIs, databases, code, browsers) and uses the results to complete a task for you.

HOW AN AGENT RUNS A TASK

Goal	›	Plan	›	Tool call	›	Observe	›	Respond
user intent		break into steps		API · SQL · search		read the result		answer or act

ANATOMY OF AN AGENT

- **Brain**: the LLM that reasons
- **Tools**: functions, MCP servers, APIs
- **Memory**: short-term context + long-term vector store
- **Planner**: decides the next step
- **Guardrails**: permissions, policies, limits

KEY PATTERNS

- **ReAct**: reason, then act, in turns
- **Function / tool calling**
- **Agentic RAG**: the agent decides what to retrieve
- **Multi-agent**: a supervisor routes work to specialist agents

FRAMEWORKS & PLATFORMS

Claude Agent SDK · LangGraph · Bedrock Agents · Azure AI Foundry · Google ADK · MCP

ACROSS DOMAINS

- **Real estate**: lead qualifier that books viewings
- **Logistics**: re-routes shipments when there's a delay
- **Finance**: KYC document checker

LIMIT → NEXT — One pass through plan → act can fail on complex work. The agent has to **check its own output and try again** until the goal is met.

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01 AUTOMATION · 02 TRAD. ML · 03 GEN AI · 04 AGENTS · 05 LOOPS · 06 FUTURE

ERA 05 / 06

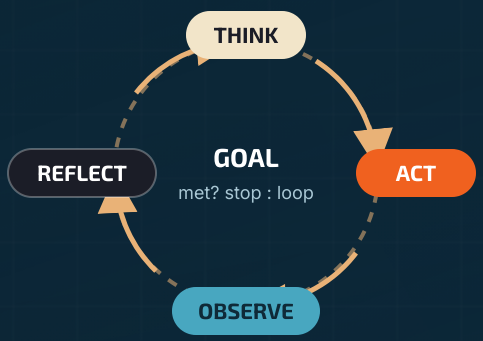
Agentic Loops

2024 → today

Agentic Loops: think · act · repeat

The agent runs in a loop. It acts, checks the result against the goal, reflects, and goes again until the job is done, the budget runs out, or a human steps in.

THE LOOP



- **Reflection**: the agent critiques its own draft
- **Plan-and-execute**: plan once, run the steps, re-plan when something fails
- **Evaluator–optimizer**: one model writes, another grades it
- **Human-in-the-loop**: a person approves risky actions
- **Stop conditions**: goal met, max steps, or token/cost budget used up

ORCHESTRATION

- State machines & graphs (LangGraph)
- Subagents with their own context
- Parallel workers, then merge the results
- Long-running, resumable runs

ACROSS DOMAINS

- **Software**: write code → run tests → fix
- **Finance**: reconcile ledgers until they balance
- **Retail**: tune pricing, measure, adjust
- **Research**: search, read, cite, verify

CONTROLS THAT MAKE LOOPS SAFE

Token & cost budgets · Tracing & observability · Permission boundaries · Sandboxed execution · Evals & regression tests · Audit logs

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ERA 06 / 06

The Future

tomorrow →

The Future: AI as a colleague

Each era built on the one before it. The next one is agents that work for hours or days, cooperate with each other, and operate in the physical world, under strong governance.

TRENDS TO WATCH

- **Long-horizon agents**: multi-day autonomous work
- **Multi-agent organisations**: teams of specialist agents
- **Computer use**: agents that operate apps and browsers
- **Embodied AI**: robotics guided by foundation models
- **Small on-device models**: private and low-latency

STANDARDS & TRUST

- **Open protocols**: MCP, agent-to-agent (A2A)
- **Governance**: data lineage, Purview, Unity Catalog
- **Regulation**: EU AI Act, model risk management
- **Safety**: evals, red-teaming, alignment

THE FULL JOURNEY

Automation	ML	GenAI	Agents	Loops	Future
humans write rules	machines learn rules	machines create	machines act	machines self-correct	machines collaborate

SKILLS FOR THE AI DEVELOPER

Python · SQL & data engineering · Prompt & context engineering · RAG & vector DBs · Agent frameworks · MCP tool building · Evals & observability · AWS · Azure · GCP AI · Data governance

START HERE Begin your AI developer journey with **Tinitiate. github.com/venkata-bcs · venkat@tinitiate.com**

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